

Use this Statement of Work document to purchase an item and/or item-related services. ⓘ

## TITLE: Physical Property Measurement System Service ⓘ

Requesting Lab/Division/Group: PML/686/686.01

Requisition Number: [to be entered by PR preparer]

### Statement of Need ⓘ

Cryocooler maintenance is critical for the Quantum Design USA Dynacool system (rm 1-3325) which is used by multiple groups (M. Stephens, K. Silverman, M. Stevens). The system is used by the various groups to validate and verify material properties of devices made in the Boulder microfabrication facility (Faint Photonics: verify transition temperatures of superconductors and evaluate single photon detectors; Sources and Detectors: verify operation of superconductors). Timely maintenance is required so that projects in these groups will not be stalled. Maintenance of the QDUSA system must be performed by the vendor. This maintenance is meant to restore the cooling power of the cryocooler by exchanging it for a refurbished unit.

### Items and/or item-related services ⓘ

| Section ⓘ                                                                   | Requirements ⓘ                                                                                                                                                                                                                                                                                                                                                    | Related Information ⓘ                                                                                               |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Items/Service</b><br>(required) ⓘ                                        | Dynacool 20K maintenance including:<br>exchange cryocooler set<br>Travel expenses and labor for site visit from vendor for maintenance service.<br><input checked="" type="checkbox"/> See attached document for detailed specifications ⓘ                                                                                                                        | Reference estimate from Quantum Design #D-197-BP-20260108, 1/8/2026                                                 |
| <b>Shipping &amp; Delivery</b> ⓘ<br><input type="checkbox"/> N/A            | Delivery information by item<br><input checked="" type="checkbox"/> Phased delivery is acceptable<br><input type="checkbox"/> Software on USB / disk required<br><input type="checkbox"/> Software electronic delivery acceptable<br>Email address for software delivery                                                                                          | ⓘ<br>Replacement parts for this service may be shipped before the vendor engineer arrives on-site for installation. |
| <b>Installation</b> ⓘ<br><input type="checkbox"/> N/A                       | Installation requirements<br><input checked="" type="checkbox"/> Site visit recommended                                                                                                                                                                                                                                                                           | ⓘ<br>Travel Expenses and Labor to be included in vendor quote.                                                      |
| <b>Inspection and Acceptance</b><br><input checked="" type="checkbox"/> N/A | Inspection requirements<br><input type="checkbox"/> Government acceptance expected to take longer than 7 days                                                                                                                                                                                                                                                     | Enter related information here                                                                                      |
| <b>Warranty</b> ⓘ<br><input type="checkbox"/> N/A                           | Standard 6 month warranty on refurbished parts.<br>If you require an Extended Manufacturer Warranty, describe the warranty requirements.<br>Period Length:Ex. month, 6 month, year etc.<br><input type="checkbox"/> Option Period 1 <input type="checkbox"/> Option Period 2<br><input type="checkbox"/> Option Period 3 <input type="checkbox"/> Option Period 4 | Includes 6 month warranty                                                                                           |
| <b>Training</b>                                                             | N/A                                                                                                                                                                                                                                                                                                                                                               | Enter related information here                                                                                      |

☒ N/A

**Maintenance**



☐ N/A

Cryocooler Cold head service

☐ See attached document for detailed maintenance specifications

Period Length: 20,000 operating hours

☐ Option Period 1

☐ Option Period 2

☐ Option Period 3

☐ Option Period 4

Cryocooler cold head 20,000 hour maintenance for Dynacool S/N D197.

**Repair**

☒ N/A

Choose an item.

Describe your repair requirements.

Period Length: Ex. month, 6 month, year etc.

☐ Option Period 1

☐ Option Period 2

☐ Option Period 3

☐ Option Period 4

Enter related information here

**Government  
Furnished  
Material /  
Equipment**

☒ N/A

List any government furnished material/equipment here

Enter related information here

**Travel**

☐ N/A

Engineer to travel to NIST for service.

*Vendor: To the maximum extent possible, include as part of appropriate sections above (maintenance, repair, etc.). Otherwise, it will be a separate line item, reimbursable per the Federal Travel Regulation.*

Enter related information here